

SIMATS ENGINEERING
C03-ASSESSMENT 1
ENUMERATION ANALYSIS
ITA1407-ETHICAL HACKING
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COURSE OUTCOME COVERED

CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)

Assessment Tool Weightage: 40%

1. scanme.nmap.org — Nmap Enumeration

Step 1: Host discovery

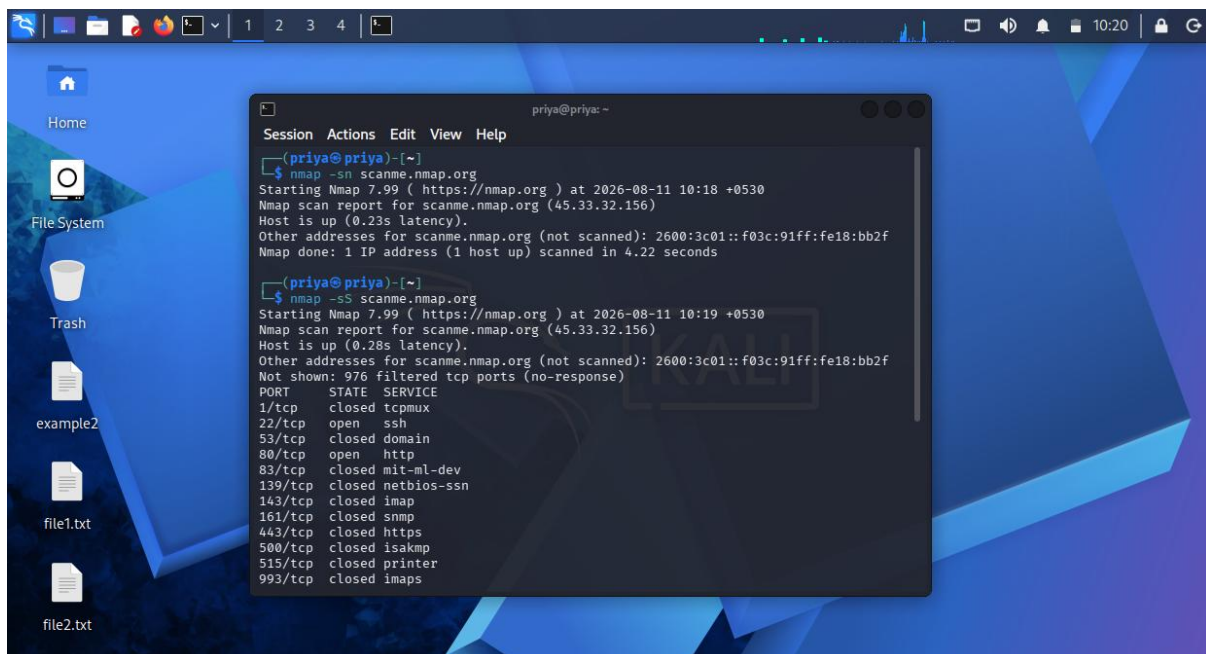
Bash

`nmap -sn scanme.nmap.org`

Step 2: TCP port scanning

Bash

`nmap -sS scanme.nmap.org`



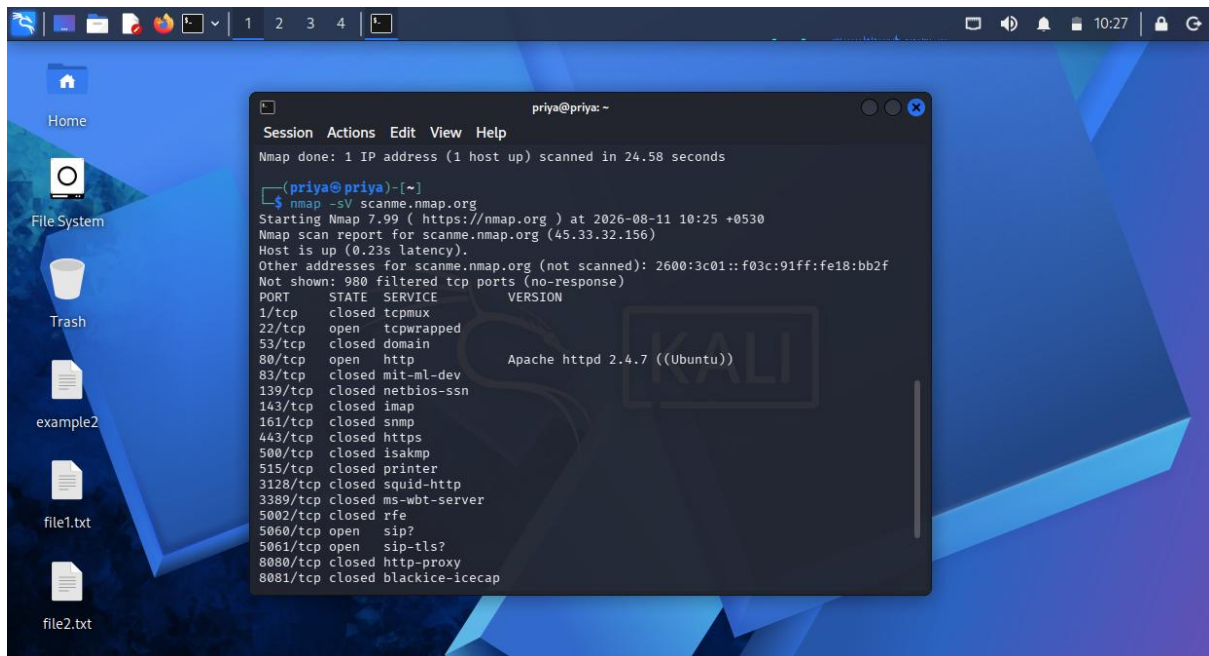
The screenshot shows a Linux desktop with a blue background. A terminal window is open, displaying the results of two Nmap scans. The first scan is a host discovery scan using `nmap -sn scanme.nmap.org`, which shows the host is up with a latency of 0.23s. The second scan is a TCP port scan using `nmap -sS scanme.nmap.org`, which shows the host is up with a latency of 0.28s and lists the open ports: 22/tcp (ssh), 80/tcp (http), and 443/tcp (https). The terminal window has a title bar that reads 'priya@priya: ~'.

```
priya@priya: ~  
Session Actions Edit View Help  
[priya@priya]~  
$ nmap -sn scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:18 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.23s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Nmap done: 1 IP address (1 host up) scanned in 4.22 seconds  
  
[priya@priya]~  
$ nmap -sS scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:19 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.28s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 976 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
443/tcp   closed https  
500/tcp   closed isakmp  
515/tcp   closed printer  
993/tcp   closed imaps
```

Step 3: Service and version detection

Bash

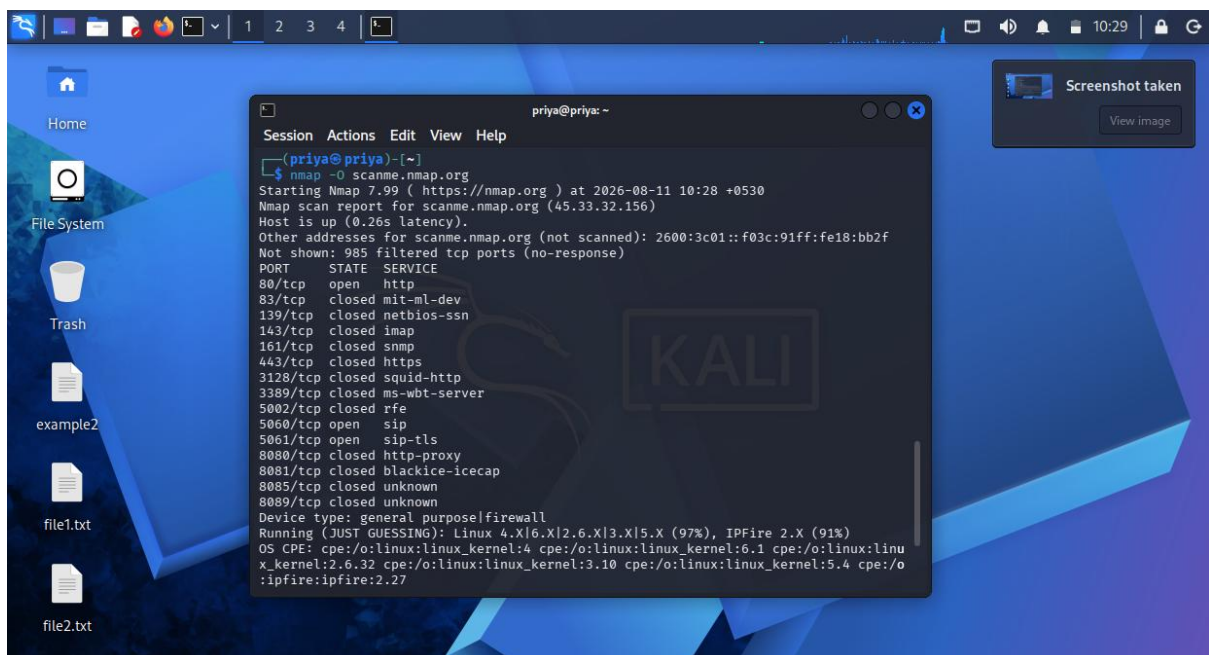
```
nmap -sV scanme.nmap.org
```



Step 4: Operating system detection

Bash

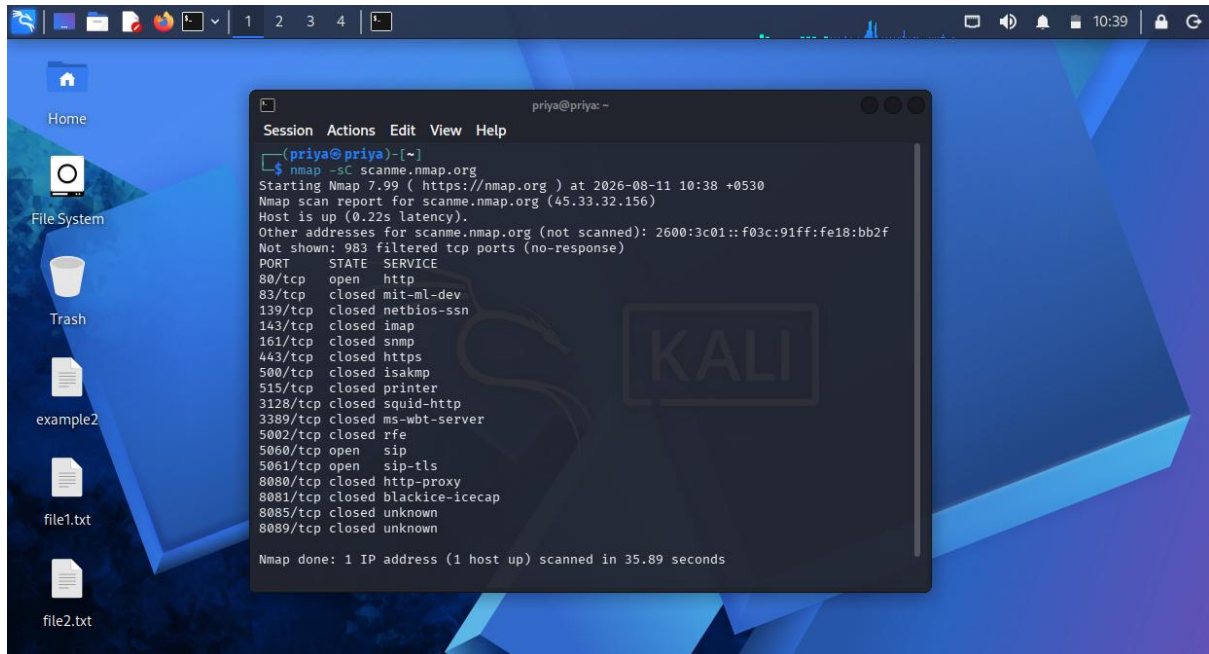
```
sudo nmap -O scanme.nmap.org
```



Step 5: Default script scanning

Bash

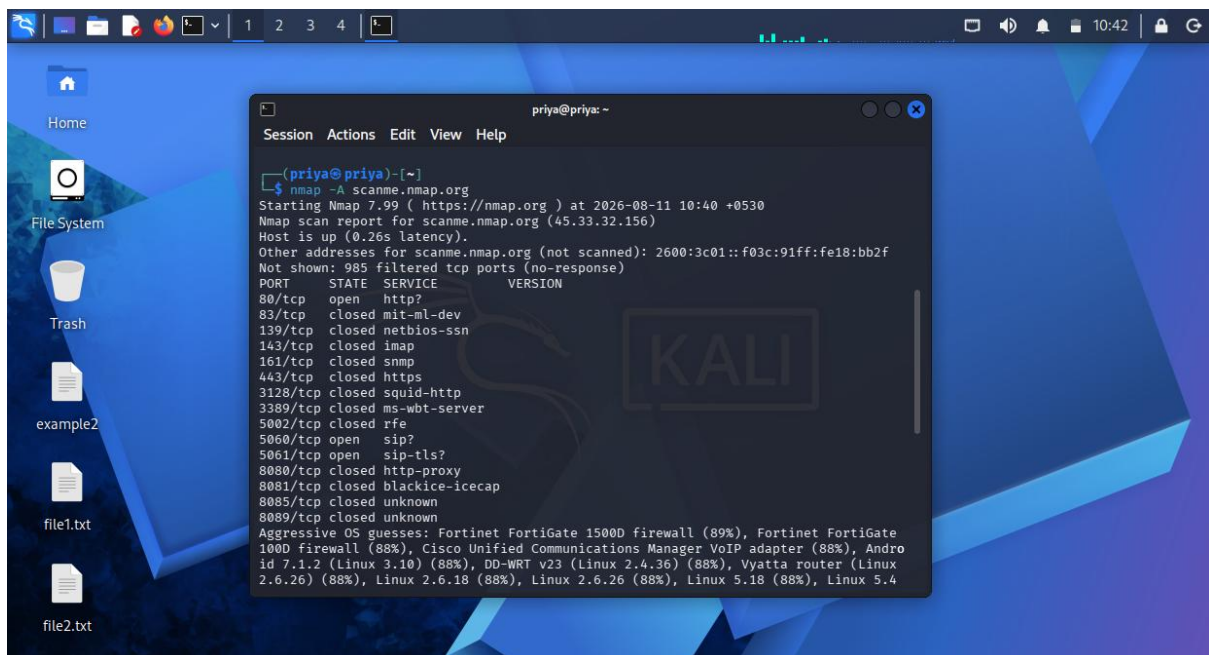
```
nmap -sC scanme.nmap.org
```



Step 6: Complete enumeration

Bash

```
sudo nmap -A scanme.nmap.org
```



2. Metasploitable 2 — 192.168.56.101

Step 1: Check whether host is active

Bash

```
nmap -sn 192.168.56.101
```

Step 2: TCP port scan

Bash

```
nmap -sS 192.168.56.102
```

Step 3: All TCP ports

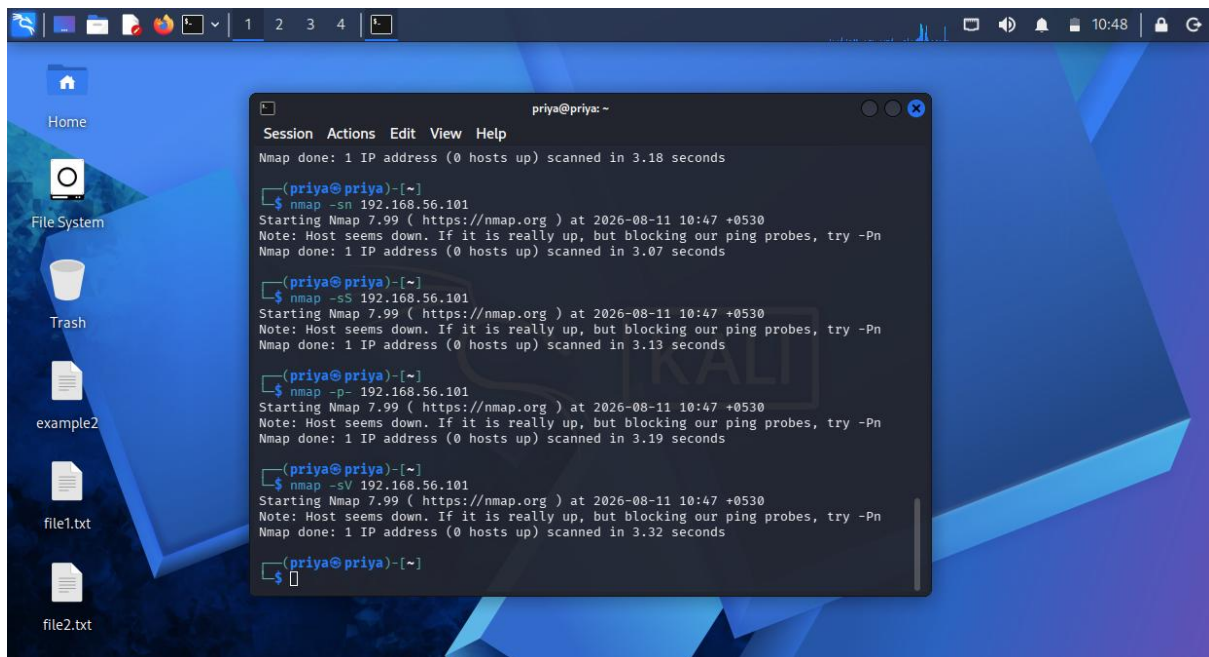
Bash

```
nmap -p- 192.168.56.102
```

Step 4: Service detection

Bash

```
nmap -sV 192.168.56.102
```



Step 5: OS detection

Bash

```
sudo nmap -O 192.168.56.102
```

Step 6: Default scripts

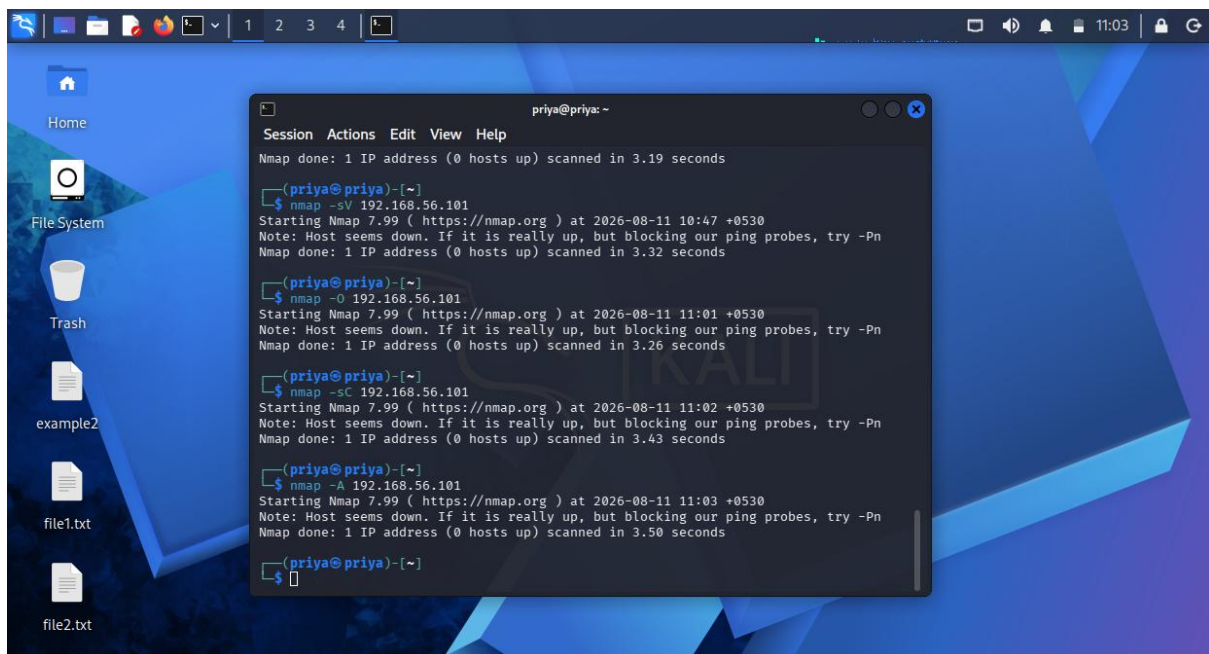
Bash

```
nmap -sC 192.168.56.102
```

Step 7: Comprehensive scan

Bash

```
sudo nmap -A 192.168.56.102
```



3. Windows Workstation — 192.168.56.102

Step 1: Host discovery

Bash

```
nmap -sn 192.168.56.102
```

Step 2: TCP port scan

Bash

```
nmap -sS 192.168.56.102
```

Step 3: All TCP ports

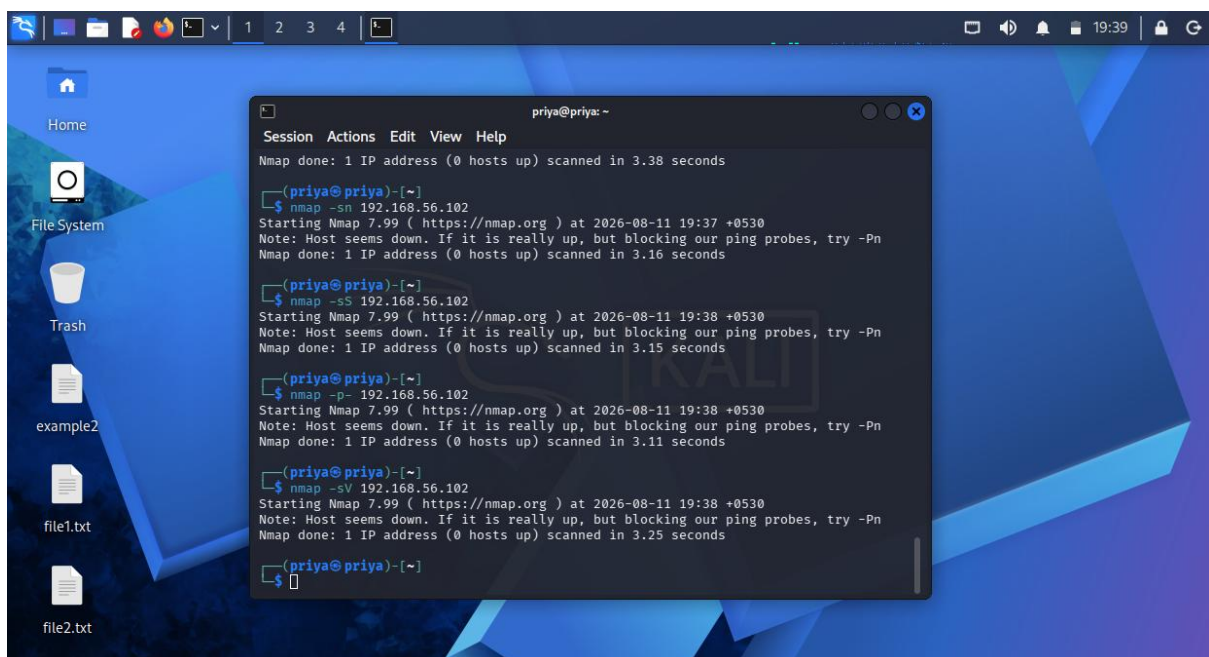
Bash

```
nmap -p- 192.168.56.102
```

Step 4: Service detection

Bash

```
nmap -sV 192.168.56.102
```



4. OWASP Juice Shop Server — 192.168.56.103

Step 1: Host discovery

Bash

```
nmap -sn 192.168.56.103
```

Step 2: Port scanning

Bash

```
nmap -sS 192.168.56.103
```

Step 3: All TCP ports

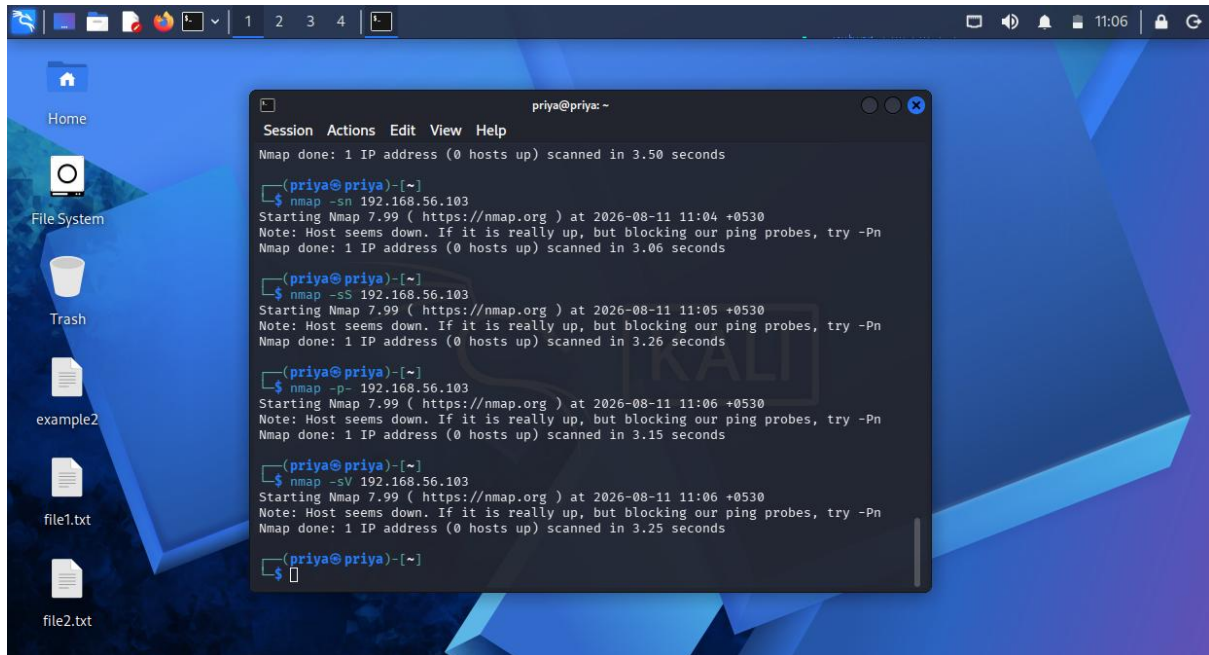
Bash

```
nmap -p- 192.168.56.103
```

Step 4: Service/version detection

Bash

```
nmap -sV 192.168.56.103
```



Step 5: OS detection

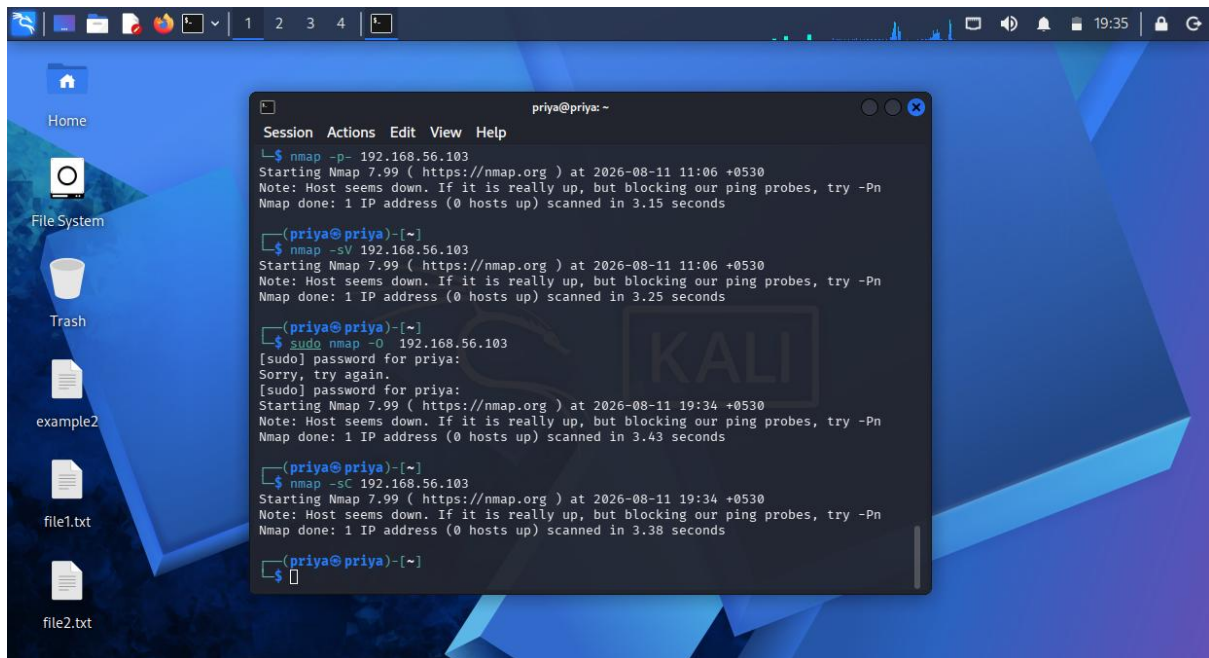
Bash

```
sudo nmap -O 192.168.56.103
```

Step 6: Default NSE scripts

Bash

```
nmap -sC 192.168.56.103
```



5. Gateway — 192.168.56.1

Step 1: Host discovery

Bash

```
nmap -sn 192.168.56.1
```

Step 2: TCP port scan

Bash

```
nmap -sS 192.168.56.1
```

Step 3: Scan all TCP ports

Bash

```
sudo nmap -p- 192.168.56.1
```

Step 4: Service detection

Bash

```
nmap -sV 192.168.56.1
```